D07 · Warehouse KPIs, Metrics & Governance

03 · Warehouse cost management & continuous improvement

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1 WHAT IT IS

Warehouse cost management tracks the cost structure — labor, building, equipment, energy, materials, technology — and drives it down through continuous improvement cycles (PDCA, kaizen) measured in cost per order.

2 WHY IT MATTERS

Labor dominates the cost of a manual warehouse, and most of its waste is travel, waiting and double handling. **The highest-return improvements are usually the simplest — slotting, layout, staffing and process — and they require no technology investment.** Cost per order integrates them all into one number that improves only if the program is real.

3 HOW IT IS DONE

1. **Break down the cost structure.** Know the share of labor, space, equipment, energy, materials and technology.
2. **Pick improvements with the PDCA cycle.** Identify, implement, measure and standardize.
3. **Run a kaizen calendar.** Several events per year, each with a baseline and documented result.
4. **Track cost per order year on year.** The integrating metric for whether improvement is happening.

IN PRACTICE

Warehouse continuous improvement is a process — not a project

A 3-day kaizen that improves LPPH 20% is a one-time success. A program that runs 4–6 kaizens per year and reviews KPIs monthly generates sustainable, cumulative improvements.

Cost per order is the financial metric that integrates all warehouse operational improvements

Improved LPPH, improved IA, and reduced overtime all reflect in cost per order.

The highest-impact warehouse kaizens are the simplest: layout, slotting, and process

Layout, slotting, and process kaizens generate 15–30% improvements without technology investment.

Connect the warehouse continuous improvement program with the S&OP

Warehouse capacity and productivity improvements must be reflected in the S&OP plan.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Continuous improvement program — distribution DC, 12-month program — 4 key initiatives implemented in sequence with monthly KPI reviews.

Initiative	Month	Measured result
Zone A re-slotting (top 200 SKUs)	Month 2	LPPH: 64 -> 82 (+28%) in Zone A
Daily cycle count for A SKUs	Month 3	IA: 96.8% -> 99.1% (+2.3 pp)
Staffing model update with actual volume	Month 4	Overtime: 18% -> 7% of total paid hours
Packing station kaizen (optimized layout)	Month 6	Packing time per order: 4.8 min -> 3.2 min (-33%)

Result: KPIs at program end (month 12): LPPH: 64 -> 84 (+31%). IA: 96.8% -> 99.2% (+2.4 pp). Cost per order: $9.80 -> $7.40 (-24%). Annual recurring savings: $480K with $45K investment.

HOW IT IS MEASURED

Cost per Order Improvement Rate % (year-on-year cost per order improvement)

Cost per Order Improvement Rate %(Prior year cost per order - Current year cost per order) / Prior year cost per order × 100

Benchmark: >5% cost per order improvement year-on-year in an active continuous improvement program A cost per order that doesn't improve year-on-year (or grows) in a stable volume environment indicates the continuous improvement program is not generating real efficiencies.

Kaizen Events per Year

Kaizen Events per YearNumber of kaizen events completed with documented results per year

Benchmark: 4–6 kaizen events per year in warehouses with active continuous improvement programs Kaizen events are the tool that materializes continuous improvement into measurable results. A warehouse with 0 kaizens per year only improves by chance — not by design.

WHAT YOU WOULD USE

Continuous Improvement

Kaizen Institute / Lean Enterprise Institute — Module: Lean Warehouse TransformationThe reference institutes for applying Lean Manufacturing and kaizen in warehouse operations.

SAP EWM Analytics + Power BI — Module: Warehouse Performance TrackingThe combination of SAP EWM reports with Power BI dashboards for continuous improvement program tracking.

4 SO WHAT

Make improvement a routine and cost per order its scoreboard. Know your cost structure, run PDCA and kaizen on a calendar, and prove results year over year.

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